

Module 23: Generative AI

Glossary

Generative AI

A subset of artificial intelligence that creates new data or content, such as text, images, music, or even entire 3D models, based on existing data

Neural Networks

Computing systems inspired by the human brain, consisting of layers of nodes (neurons) that learn to recognize patterns in data through training

Recurrent Neural Networks (RNNs)

A type of neural network that works well for machine learning problems involving sequential data

Generative Adversarial Networks (GANs)

A type of neural network framework consisting of two models, a generator and a discriminator, that compete against each other to improve their performance

Generator

A neural network in a GAN that generates new data by learning to mimic the distribution of real data (e.g., generating realistic images)

Discriminator

The neural network in a GAN that distinguishes between real data and data generated by the generator, helping improve the generator's performance

Diffusion Models

A type of generative model that creates data by reversing a process that gradually adds noise to a starting point, eventually arriving at a final structured output

De-noising

The process of removing noise from an image or data point to generate the final, clear output. Diffusion models are designed to perform denoising effectively

Transformer-Based Models

A type of neural network architecture designed to handle sequences of data, like text, by considering the relationships between elements in the sequence. Transformers are the foundation for models like ChatGPT

Embedding

A technique used to convert tokens (words or sub-words) into fixed-length vectors in a high-dimensional space, where similar tokens are placed closer together based on their meanings

Unembedding

The final phase in transformer models, where an enriched embedding is used to predict the next token in a sequence, converting the high-dimensional vector back into a token from the vocabulary

N-Gram Model

A sequence of 'n' items (words, letters, etc.) from a given text, often used in language modeling to predict the next item in a sequence

Vanishing Gradient Failure

A problem in training deep neural networks where gradients become very small, making it difficult to update weights and learn long-range dependencies

Mode Collapse Failure

A common issue in GANs where the generator produces only a limited variety of outputs, failing to capture the diversity in the training data

Tokenization

The process of breaking down text into smaller units called tokens, such as words or sub-words

Self-Attention Mechanism

A key feature of transformers that allows tokens in a sequence to "attend" to or communicate with each other, refining their meanings based on the context of surrounding tokens

Stochastic Gradient Descent (SGD)

A popular optimization algorithm used to train models by iteratively updating the weights based on the gradient of the loss function with respect to each training example

Temperature

A setting in text generation models that controls the randomness or creativity of outputs. Lower temperatures result in more predictable outputs, while higher temperatures produce more diverse results